

Unit - IV

Push Down Automata (PDA)

Introduction:

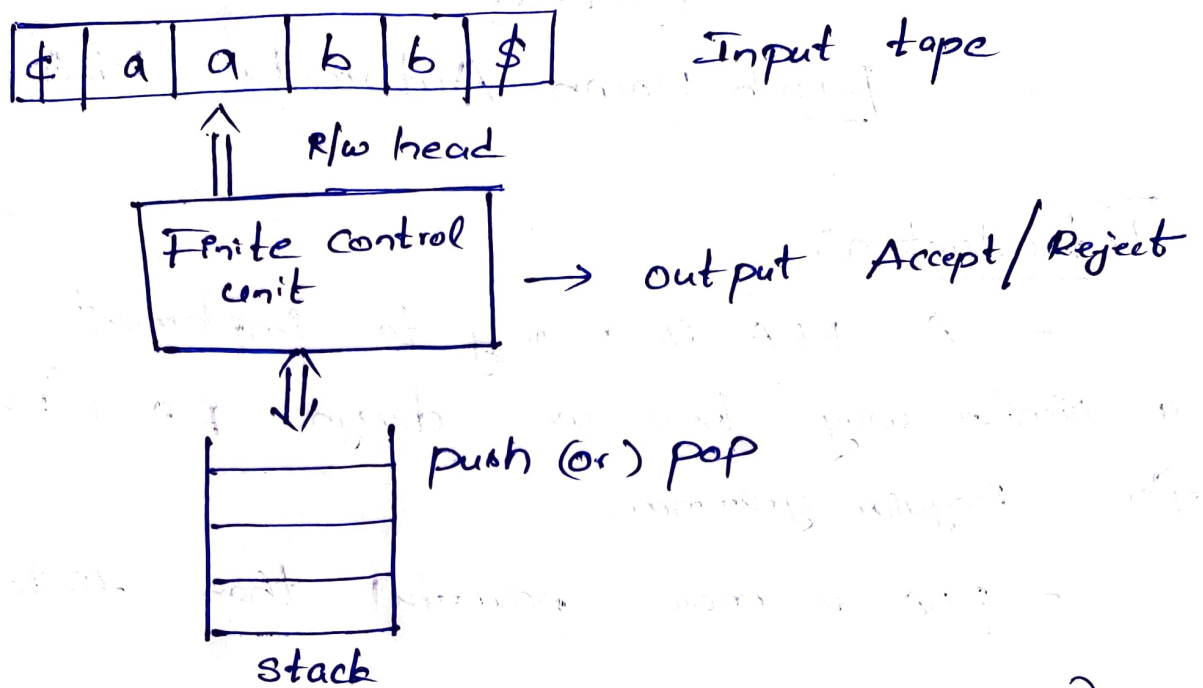
- A PDA is a way to implement a CFG in a similar way how we design F.A (Finite Automata) for Regular grammar.
- PDA is more powerful than finite state Machine (FSM)
- Finite state machine has a very limited memory. But PDA has more memory
- Simply PDA can be defined as combination of finite state machine (FSM) and stack.

$$\text{PDA} = \text{FSM} + \text{stack}$$

Model of PDA (or) Block diagram of PDA:

- PDA has mainly three components

- i) Input tape
- ii) Finite control unit
- iii) Stack



Model of push down Automata (PDA)

i) Input Tape :

- Input tape is a place where input symbols are hold, which are going to be processed.

ii) Finite control unit :

- Finite control unit will takes the input symbols from input tape and perform certain operations like transition function.

- It also will takes the stack operations result's and decide the processed string is accepted (✓) or rejected (✗).

iii) Stack :

- It will performs operation like push (or) pop and gives result to Finite control unit.

Definition of Push Down Automata :

Mathematically PDA can be defined by using 7-tuple. They are :

$$M = \{ Q, \Sigma, \Gamma, \delta, q_0, Z_0, F \}, \text{ where}$$

Q - finite and non-empty set of states

Σ - finite and non-empty set of input symbols called an alphabet

Γ - finite and non-empty set of stack symbols

δ - Transition function

q_0 - Initial state

Z_0 - Top element/symbol in stack

F - Final state

Transition Function (δ) :

In PDA transition function is given as follows

$$\delta: Q \times \{ \Sigma \cup \epsilon \} \times \Gamma^* \rightarrow Q \times \Gamma^*$$

$$\delta: \boxed{Q \times \Sigma^* \times \Gamma^* \rightarrow Q \times \Gamma^*}$$

- which means transition function has all possible combination's of "states, input symbols, stack symbols" as input's and gives all possible combination of "states & stack symbols" as pairs for output.

- Here ' δ ' takes 3-tuples as "input" like:

$$\delta(q, a, x)$$

- Here ' δ ' gives finite set of pairs as "output" like:

$$\{(p_1, \gamma_1), (p_2, \gamma_2), \dots, (p_m, \gamma_m)\}$$

$$\delta(q, a, x) = \{(p_1, \gamma_1), (p_2, \gamma_2), \dots, (p_m, \gamma_m)\}$$

where q - state in Q

a - Either input symbol from Σ (a) $\in$

x - Stack symbol, i.e member of τ
($\emptyset$)

Top symbol in stack

p - It is new state

γ - It is set of stack symbols that replaces ' x ', at the top of stack.

Example: Construct PDA for Language, $L = \{ a^n b^n \mid n \geq 1 \}$

Sol: Let us construct language for above condition

$$L = \{ ab, aabb, aaabbb, \dots \}$$

- we know that based on above language, there will be equal no. of a's and b's

- To construct PDA for above condition, first we need to push all a's and pop all b's then the stack will be empty. which indicates both symbols a & b are equal.

- Now consider one string from above language to construct PDA

string: aaabbb

- Now, apply PDA transition function on above input string.

$$\delta(q, a, x) = (p, \gamma)$$

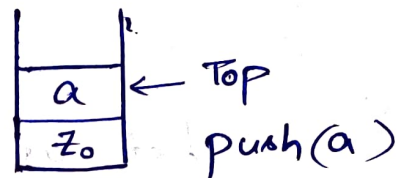
(Input)

(output)

- now consider single input symbol's from string

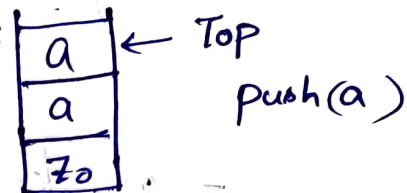
Input a:

$$\delta(q_0, a, z_0) = (q_0, az_0)$$



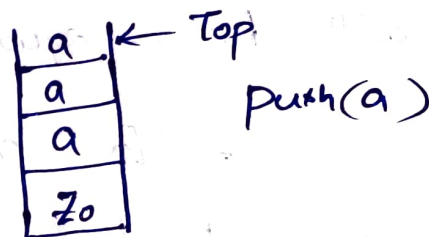
Input a:

$$\delta(q_0, a, a) = (q_0, aa)$$



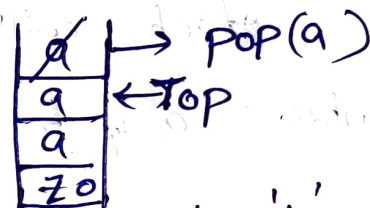
Input a:

$$\delta(q_0, a, a) = (q_0, aaa)$$



Input b:

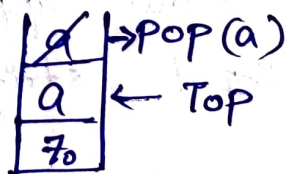
$$\delta(q_0, b, a) = (q_1, \epsilon)$$



- now we need to pop a 's when we get 'b' as input and state also will change
- In output of transition function (q_1, ϵ) means doing pop operation in input.

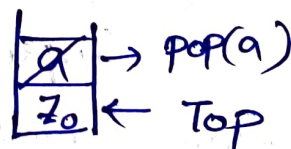
Input b:

$$\delta(q_1, b, a) = (q_1, \epsilon)$$



Input b:

$$\delta(q_1, b, a) = (q_1, \epsilon)$$



- Now no more 'b's to input and a 's to pop

$$\delta(q_1, \epsilon, z_0) = (q_2, z_0)$$



- when ' ϵ ' appears as input string in the input of transition function, then it is end of the string (& no more symbols to process).
- Now we can define PDA as follows:

$$M = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$$

$$Q = \{q_0, q_1, q_2\}$$

$$\Sigma = \{a, b\}$$

$$\Gamma = \{a, z_0\}$$

$$q_0 = \{q_0\}$$

$$z_0 = \{z_0\}$$

$$F = \{q_2\}$$

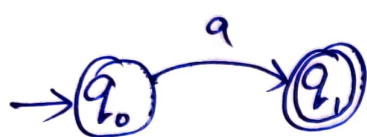
now

$$M = \{ \{q_0, q_1, q_2\}, \{a, b\}, \{a, z_0\}, \delta, \{q_0\}, \{z_0\}, \{q_2\} \}$$

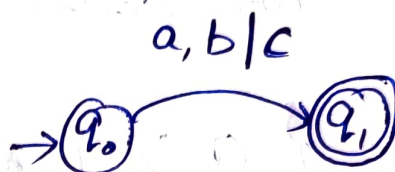
Graphical notation of PDA :

- Graphically PDA can be represented by "transition diagram".

In Finite Automata



In PDA



where a - input symbol

b - Top of Stack

c - new top of stack.

Example : $L = \{ a^n b^n \mid n \geq 1 \}$

PDA : $\delta(q_0, a, z_0) = (q_0, a z_0)$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, a, a) = (q_0, aaa)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, z_0)$$

Graphical representation of PDA

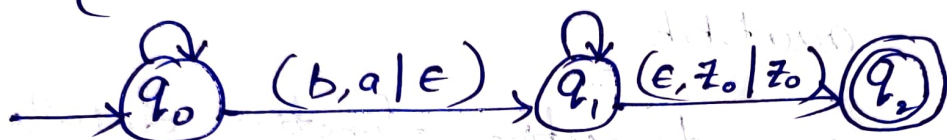
(8)

Transition diagram of PDA

$(a, a | aa)$

$(a, z_0 | az_0)$

$(b, a | \epsilon)$



Instantaneous Description (ID) :

- ID is an informal notation that explains how a PDA computes the given string.
- Generally ID is used for acceptance of string by PDA or not.
- ID can be represented by triple (q, w, α)

where

q - Current state

w - Input string

α - current content of stack.

- ID also represented by :

$\vdash$ (Fallen T) : Turnstile notation

$\vdash$ (Single move)

$\vdash^*$ Multiple / Sequence of moves

Example :

$$L = \{ a^n b^n \mid n \geq 1 \}$$

$$L = \{ ab, aabb, aaabbb, \dots \}$$

- Now consider any string from above language
for accepted or not by PDA

- String : aaabbb

- Now write all transition function details and

create ID for it.

Transition Function :

$$\delta(q_0, a, z_0) = (q_0, az_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, a, a) = (q_0, aaa)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, z_0)$$

Instantaneous Description

$$ID = (q, w, \gamma)$$

$$\begin{array}{c} (q_0, aaabbb, z_0) \vdash (q_0, aaabbb, az_0) \\ \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \end{array}$$

$$\vdash (q_0, abbb, aaaz_0)$$

$$\vdash (q_0, bbb, aaaaz_0)$$

$$\vdash (q_1, bb, aaaz_0)$$

$$\vdash (q_1, b, aaaz_0)$$

$$\vdash (q_1, \epsilon, aaaz_0)$$

$$\vdash (q_2, \epsilon)$$

$$\boxed{q_2, \epsilon}$$

No elements / symbols present

in stack, so string is "Accepted"

PDA Accepting languages :

- The string / Language acceptency in PDA can be done in two ways :

- i) Final state Acceptency
- ii) Empty stack Acceptency.

Final state Acceptance :

For PDA, $M = (Q, \Sigma, \Gamma, \delta, Z_0, F)$ we define $L(M)$, the language accepted by 'final state' to be:

$$\{ w \mid (q_0, w, \epsilon) \vdash^* (p, \epsilon, \gamma) \text{ for } p \in F \text{ and } \gamma \in \Gamma^* \}$$

Empty state Acceptance :

For PDA, M , the language accepted by "empty stack" to be:

$$\{ w \mid (q_0, w, \epsilon) \vdash^* (p, \epsilon, \epsilon) \text{ for some } p \in Q \}$$

Equivalence of DFA & CFG :

- i) CFG to PDA conversion
- ii) PDA to CFG conversion

CFG to PDA Conversion :

- There are certain steps to be followed for conversion. They are:

Step 1: Grammar must be in GNF

Step 2: push 'S' on to stack

$$\delta(q_0, \epsilon, z_0) = (q_1, S z_0)$$

Step 3: For production

Non Terminal (NT_1) $\rightarrow$ Single Terminal (T) { string of Non-terminals (NT) }

$$\delta(q_1, T, NT_1) = (q_1, \text{string of NT})$$

Step 4:

For the production

$$NT_1 \rightarrow \{ \text{Single T} \}$$

$$\delta(q_1, T, NT_1) = (q_1, \epsilon)$$

Example: Construct PDA for the grammar

$$S \rightarrow aB$$

$$B \rightarrow bA \mid b$$

$$A \rightarrow aB$$

Sol: Step ①: Given grammar is already in GNF, no need for conversion

Step ②: push 's' on to the stack

$$\delta(q_0, \epsilon, z_0) = (q_1, Sz_0) \quad \begin{array}{|c|} \hline S \\ \hline z_0 \\ \hline \end{array}$$

Step ③ & ④: Implement the productions PDA for given productions

$$S \rightarrow aB$$

which is form $\delta(NT_i \rightarrow T NT)$

$$\delta(q_1, a, S) = (q_2, B) \quad \begin{array}{|c|} \hline S \\ \hline z_0 \\ \hline \end{array} = \begin{array}{|c|} \hline B \\ \hline z_0 \\ \hline \end{array}$$

$$B \rightarrow bA$$

$$\delta(q_2, b, B) = (q_3, A) \quad \begin{array}{|c|} \hline B \\ \hline z_0 \\ \hline \end{array} = \begin{array}{|c|} \hline A \\ \hline z_0 \\ \hline \end{array}$$

which is form $\delta(NT_i \rightarrow T) B \rightarrow b$

$$\delta(q_3, b, B) = (q_4, \epsilon) \quad \begin{array}{|c|} \hline B \\ \hline z_0 \\ \hline \end{array} = \begin{array}{|c|} \hline z_0 \\ \hline \end{array}$$

$$A \rightarrow aB$$

$$\delta(q_4, a, A) = (q_5, B) \quad \begin{array}{|c|} \hline A \\ \hline z_0 \\ \hline \end{array} = \begin{array}{|c|} \hline B \\ \hline z_0 \\ \hline \end{array}$$

- Now finally obtained PDA from CFG is:

$$\text{CFG} \begin{cases} S \rightarrow AB \\ B \rightarrow bA \mid b \\ A \rightarrow aB \end{cases} \Rightarrow \text{PDA} \begin{cases} \delta(q_0, \epsilon, z_0) = (q_1, Sz_0) \\ \delta(q_1, a, S) = (q_1, B) \\ \delta(q_1, b, B) = (q_1, A) \\ \delta(q_1, b, B) = (q_1, \epsilon) \\ \delta(q_1, a, A) = (q_1, B) \end{cases}$$

- Now let us check whether the obtained PDA is correct (or) not

- For this let us consider string from given grammar and apply them on PDA transitions.

String : abab

<u>input s:</u>	$(q_0, \epsilon, z_0) = (q_1, Sz_0)$	} verify them in PDA transitions
<u>input a:</u>	$(q_1, a, S) = (q_1, B)$	
<u>input b:</u>	$(q_1, b, B) = (q_1, A)$	
<u>input a:</u>	$(q_1, a, A) = (q_1, B)$	
<u>input b:</u>	$(q_1, b, B) = (q_1, \epsilon)$	
<u>input ϵ:</u>	$(q_1, \epsilon, z_0) = (q_2, z_0)$	

(Acceptance of String by final state)

$$if \quad \delta(q_1, \epsilon, z_0) = (q_2, \epsilon)$$

(Acceptance of string by Empty stack)

Now taken string is accepted.

PDA to CFG Conversion:

Example:

Convert the PDA to CFG

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, b, a) = (q_1, a)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_1, \epsilon)$$

Sol:

$$PDA(P) = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$$

$$= (\{q_0, q_1\}, \{a, b\}, \{a, z_0\}, \delta, q_0, z_0, F)$$

Now Construct CFG:

$$G = (V, T, P, S)$$

Variable(V):

$$\text{where } V = \left\{ S, [q_0 a q_0], [q_0 z_0 q_0], [q_1 a q_1], [q_1 z_0 q_1], [q_0 a q_1], [q_0 z_0 q_1], [q_1 a q_0], [q_1 z_0 q_0] \right\}$$

where above combinations are taken from $\{q_0, q_1\}$ & $\{a, z_0\}$ (all possible combinations)

Terminal (T):
where $T = \{a, b\}$

production (P):

$$i) S \rightarrow [\underbrace{q_0}_{\text{initial state}}, \underbrace{z_0}_{\text{initial bp of stack}}, \underbrace{q_0}_{\text{initial state}}]$$

$$S \rightarrow [\underbrace{q_0}_{\text{initial state}}, \underbrace{z_0}_{\text{initial bp of stack}}, \underbrace{q_1}_{\text{initial state}}]$$

- R.H.S side of production obtained from initial state and top of stack along with different states. $\therefore$ where q_0, z_0 are initial state & initial bp of stack

$\therefore$ where q_0, q_1 are states

- Now create production from PDA transition:

$$ii) \delta(q_0, a, z_0) = (\underbrace{q_0, az_0}_{\text{initial state}})$$

Here we have two values
 $2^2 = 4$

$$[] \rightarrow [] - ①$$

$$[] \rightarrow [] - ②$$

$$[] \rightarrow [] - ③$$

$$[] \rightarrow [] - ④$$

if we have one value
 $2^1 = 2$

Here we have 2^2 power value = 2, then two columns of R.H.S side

$$[] \rightarrow [] []$$

$$[] \rightarrow [] []$$

$$[] \rightarrow [] []$$

$$[] \rightarrow [] []$$

Now fill the values

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\begin{aligned} [q_0 z_0 q_0] &\rightarrow a[q_0 a q_0] [q_0 z_0 q_0] \\ [q_0 z_0 q_0] &\rightarrow a[q_0 a q_1] [q_1 z_0 q_0] \\ [q_0 z_0 q_1] &\rightarrow a[q_0 a q_0] [q_0 z_0 q_1] \\ [q_0 z_0 q_1] &\rightarrow a[q_0 a q_1] [q_1 z_0 q_1] \end{aligned}$$

— where q_0, q_1 on L.H.S side based on states available

Now consider another PDA transition:

$$\delta(q_0, b, a) = (q_1, \underline{a})$$

Here we have one value

$$[] \rightarrow []$$

So $q' = q$

$$[] \rightarrow []$$

Here we have q' , power value is '1', then only one column on R.H.S side production.

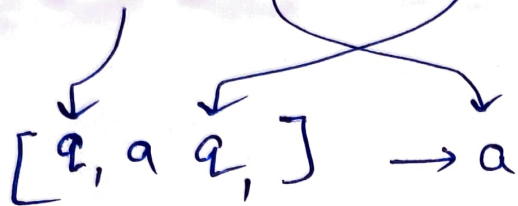
$$\delta(q_0, b, a) = (q_1, a)$$

$$[q_0 a q_0] \rightarrow [q_1 a q_0]$$

$$[q_0 a q_1] \rightarrow [q_1 a q_1]$$

fa

$$\delta(q, a, a) = (q, \epsilon)$$



Here $z^0 = 1$